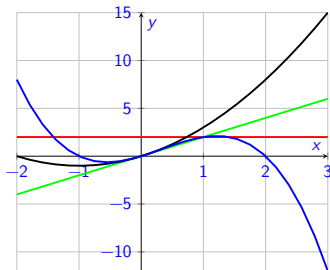


More About Functions

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The function of today's class is to learn about:

- 1 The function of today's class is to learn about:
- 2 Functions: notation
 - The real numbers
 - Domain of a function
- 3 4 Ways to Represent a Function
- 4 Graphical Representation
- 5 A Catalog of Functions
 - Linear functions
- 6 Polynomials
 - Linear
- 7 Exercises

For more, see Sections 1.1 and 1.2 of [https://math.libretexts.org/Bookshelves/Calculus/Calculus_\(OpenStax\)/01%3A_Functions_and_Graphs](https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/01%3A_Functions_and_Graphs)

$$f(x) = x^2$$

Recall: This section is all about **functions**, which a “rule” for mapping inputs to outputs.

1. Writing $f : A \rightarrow B$ means the inputs come from the set A , and the outputs come from the set B . (A **set** is just a collection of things).
2. A is called the **domain**, and B is called the **co-domain**.
3. $y = f(x)$ means “ x gets mapped to y according to the rule defined by f ”. We sometimes also say “ y is the image of x ”.
4. The subset of B that contains all the images of the things in A is called the **range** of f .
5. When we write $x \in A$ we mean “ x is an element of A , or “ x belongs to A ”.

$f(x) = x^2$ with A as the set of (real) numbers and B is also the set of reals.

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The Real Numbers, $\mathbb{R}$

In almost every example we'll consider, in MA140, f will a mapping from one **real** number to another.

The set of **real** numbers, denoted $\mathbb{R}$, are the numbers used to represent continuous quantities, such as length, speed, time and temperature.

- ▶ **continuous** means that there are no gaps, or "jumps". as a car accelerates from, say, 0m/s to 20m/s, at some point it will have taken on every speed between these values; where it has had .
- ▶ real numbers can be positive or negative.

Often, the domain of a function is not explicitly stated.
In such a case the following **Domain Convention** applies.

The **domain** of a function f is the set of all numbers x for which $f(x)$ makes sense and gives a *real-number output*.

Example

- Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \frac{1}{x^2 - x}$.

Note • $x=2$, $f(2) = \frac{1}{4-2} = \frac{1}{2}$ ✓

$x=1$ $f(1) = \frac{1}{1-1} = \frac{1}{0}$ ✗ *not defined*

so $x=1$ is not in the domain of f
because it leads to division by zero.

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In such a case the following **Domain Convention** applies.

The **domain** of a function f is the set of all numbers x for which $f(x)$ makes sense and gives a *real-number* output.

Example

- Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \frac{1}{x^2 - x}$.
 x is in the domain of f , so long as $f(x)$ "makes sense"; in this case no division by zero.

$$f_1(x) = \frac{1}{x(x-1)} \quad \begin{array}{l} x(x-1) = 0 \text{ if} \\ x = 0 \text{ or } x-1 = 0. \end{array}$$

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Example

- Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \frac{1}{x^2 - x}$.

so the domain of f_1 is
all of $\mathbb{R}$ except 0 and 1.
That is : $\mathbb{R} \setminus \{0, 1\}$.

Example

Find the subset of $\mathbb{R}$ that is the **domain** of the function

$$f_2(x) = \sqrt{x+2}.$$

Since we work with the real numbers, $\mathbb{R}$, we can't have the square root of a negative number.

$$\text{eg } f_2(2) = \sqrt{2+2} = \sqrt{4} = 2 \quad \checkmark$$

$$f_2(0) = \sqrt{0+2} = \sqrt{2} \quad \checkmark$$

$$f_2(-4) = \sqrt{-4+2} = \sqrt{-2} \quad \text{X.}$$

So x is in the domain of f if $x+2 \geq 0$

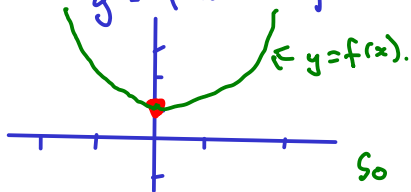
So $x \geq -2$. Same as $[-2, \infty)$

Example

Given the function $f_3(x) = 3x^2 + 1$, find the largest subset of $\mathbb{R}$ that is the domain of f_3 . What is the corresponding range?

① Since $f(x) = 3x^2 + 1$ is defined for all $x \in \mathbb{R}$, then the domain is (all of) $\mathbb{R}$

② Range : set of values y for which $y = f(x)$ for some x .



$$\begin{aligned} f(0) &= 1 \\ f(1) &= 4 \\ f(-1) &= 4 \end{aligned}$$

So range is $[1, \infty)$ i.e. $1 \leq y < \infty$

Example

Identify the domain (in $\mathbb{R}$) and range of

$$f_4(x) = \sqrt{(x+4)(3-x)}$$

For x to be in the domain of f we need

$$(x+4)(3-x) \geq 0$$

So we need either (i) $x+4 \geq 0$ and $3-x \geq 0$
or (ii) $x+4 \leq 0$ and $3-x \leq 0$.

$$(i) \quad x+4 \geq 0 \Rightarrow x \geq -4. \quad 3-x \geq 0 \Rightarrow 3 \geq x$$

So any x with $-4 \leq x \leq 3$ is in the domain.

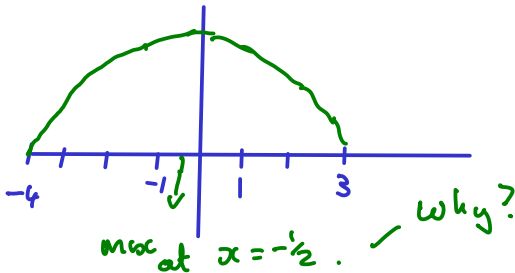
(ii) $x \leq -4$ and $x \geq 3$. Not possible!

Example

Identify the domain (in $\mathbb{R}$) and range of

$$f_4(x) = \sqrt{(x+4)(3-x)}$$

For the range we need to find all values of $f_4(x)$ for $-4 \leq x \leq 3$.



(come back in 2 weeks!)

Example

Identify the domain and range of $f_5(x) = \frac{1}{x}$.

Domain is all of $\mathbb{R}$ except $x=0$.
So the domain is $(-\infty, 0) \cup (0, \infty)$ i.e.
 $\mathbb{R} / \{0\}$. set notation!

Range:



$y = f(x)$:

so the
range too
is $\mathbb{R} / \{0\}$.

4 Ways to Represent a Function

A function can be represented in different ways:

1. **verbally** (by a description in *words*);
2. **numerically** (as a *table* of values);
3. **visually** (as a *graph*);
4. **algebraically** (by an explicit *formula*).

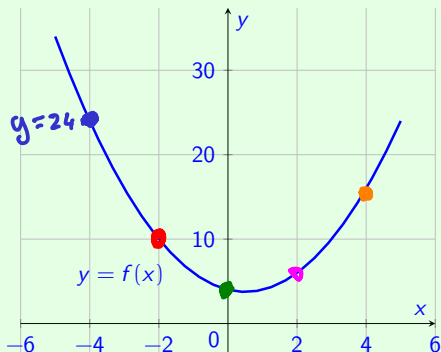
Often it is possible, and useful, to go from one way to another.

Especially from algebraic to visual.

Graphical Representation

Graph $\rightarrow$ Table

A common way to *visualize* a function $f: X \rightarrow \mathbb{R}$ is its *graph* in the x, y -plane. In this example, $f(x) = x^2 - x + 4$.



$$f(-4) = 16 - (-4) + 4 = 24$$

x	$f(x)$
-4	24
-2	10
0	4
2	6
4	16

A Catalog of Functions

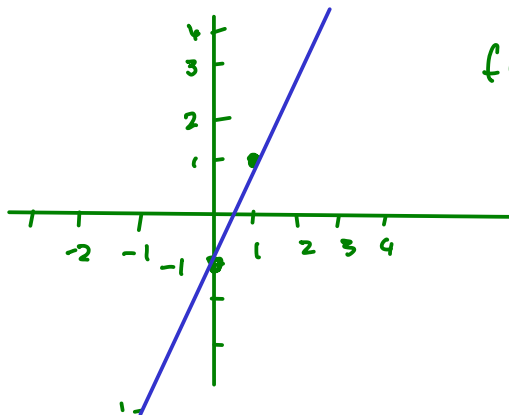
There are many *different types of functions* that can be used to *model relationships* between objects in the *real world*.

The most common types of functions (in MA140) are:

- ▶ Linear Functions, $f(x) = ax + b$
- ▶ Polynomial Functions, $x^3 + 3x^2 - 1$
- ▶ Power Functions, $f(x) = x^{-1/2}$ or $f(x) = x^{1.23}$
- ▶ Rational Functions, $f(x) = \frac{1+x}{1+x^2}$
- ▶ Algebraic Functions, $f(x) = \sqrt{x}$ or $f(x) = x^{3/2}$
- ▶ Trigonometric Functions, $f(x) = \sin(x)$
- ▶ Exponential Functions, $f(x) = e^x$
- ▶ Logarithms, $f(x) = \log(x)$

Linear functions have formulae such as $f(x) = mx + c$, where m and c are some given numbers.

It is often represented graphically as a straight line of slope m through the point $(0, c)$.



$$f(x) = 2x - 1$$

$$f(0) = -1$$

$$f(1) = 1$$

Polynomials

A **polynomial function** (or just **polynomial**) is a function of the form

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0, \quad x \in \mathbb{R},$$

where $a_0, a_1, \dots, a_n$ are real numbers called the **coefficients** of the polynomial.

The number n is called the **degree** of the polynomial.

There are special names for polynomials of low degree:

$n=0$ $f(x) = a_0$ *constant*

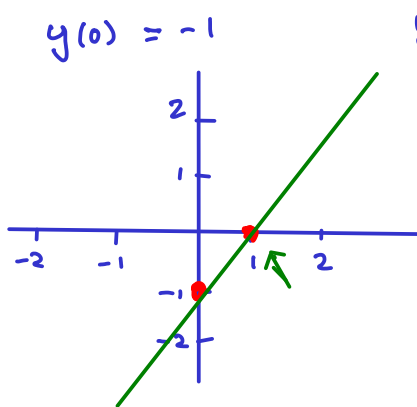
$n=1$ $f(x) = a_1 x + a_0$ *linear*

$n=2$ $f(x) = a_2 x^2 + a_1 x + a_0$ *quadratic*

$n=3$ $f(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$ *cubic*

Example: Linear Polynomial

A polynomial of degree $n = 1$ is called “linear”. Its graph is a straight line. E.g. $y = x - 1$ is a **linear** polynomial.



Finished here
Wednesday.

Exercise 1.2.1

Identify the largest possible subset of $\mathbb{R}$ that could be the domain and range of these functions:

1. $f(x) = (x - 4)^2 + 5$

2. $f(x) = \sqrt{3x + 2} - 1$

3. $f(x) = 3/(x - 2)$.

(See Example 1.1.2 of the textbook).

Exercise 1.2.2 (MA140 2025/2026 Exam)

For each of the following functions, determine the largest possible subsets of $\mathbb{R}$ that could be its domain and range.

1. $f(x) = (x - 4)^4 + 5$.
2. $f(x) = \sqrt{3x + 2} - 1$.

